Attendance & Departure System

Setup Guide

Installation, configuration and putting the system online

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Contents

1. Overview and requirements
2. Install Python
3. The system folder
4. First run
5. Secure and configure the system
6. Make it reachable from phones
7. Start automatically with Windows
8. Install the app on phones
9. Backups, logs and restore
10. Troubleshooting
11. Security checklist

1. Overview and requirements

The Attendance & Departure System has three parts:

- **Server**: one program (`server.py`) that runs on a Windows PC and keeps all data in a single database file.
- **Admin dashboard**: opened in any web browser by HR, managers and the system admin.
- **Employee app**: the same web address opened on a phone and added to the home screen. Employees check in and out with GPS.

What you need

Item	Requirement
Server PC	Windows 10 or 11, always on while the system is in use, with a stable internet connection.
Python	Version 3.10 or newer. No other packages are needed.
Phones	Android (Chrome) or iPhone (Safari), with GPS turned on.
HTTPS	Phones only allow GPS on a secure `https://` address. Chapter 6 explains how to get one.
Domain (optional)	For a permanent internet address, e.g. `attendance.yourcompany.com`, about USD 10 per year.

Setup at a glance

Install Python → Copy the system folder → First run → Secure the admin account → Configure → Go online → Install on phones

2. Install Python

1. Download Python 3 for Windows from `https://www.python.org/downloads/` .
2. Run the installer and tick **Add python.exe to PATH** on the first screen, then click **Install Now**.
3. Open **PowerShell** and check the installation:

```
py --version
```

You should see `Python 3.10` or newer.

3. The system folder

Copy the whole system folder to the server PC, for example `C:\in-out` . Keep the folder structure as it is.

File / folder	Purpose
`server.py`	The server program.

File / folder	Purpose
`run.bat`	Double-click to start the server in a window.
`online.ps1`	Puts the system on the internet through Cloudflare Tunnel.
`autostart.ps1`	Starts the server automatically with Windows.
`gen_cert.bat`	Creates a certificate for use on the local network only.
`static\`	The web app (dashboard and phone app).
`data\`	Created on first run: the database `attendance.db`, the `backups\` folder and `server.log`.

> **Protect the data folder**
> The `data` folder holds every employee record and attendance punch. Do not share it, email it or upload it to GitHub, and include it in your regular PC backups.

4. First run

Option A: try it with demo data

Double-click `run.bat`, or run in PowerShell inside the folder:

```
py server.py
```

On first start the system creates demo data: 12 employees, 3 shifts, 2 locations and 5 weeks of attendance. Open `http://localhost:8000` on the server PC.

Role	Username	Password
System admin	`admin`	`Admin@123`
HR	`hr`	`Hr@12345`
Line manager	`manager`	`Manager@123`
Employee	`1002` … `1012`	`Emp@1234`

Option B: start clean for real use

To delete the demo data and start with only an admin account, run this once in a console window:

```
py server.py --reset-demo --no-demo
```

The console prints the new admin password, for example `username: admin password: Adm1a2b3c4d1`. Write it down. To choose the password yourself:

```
$env:ATT_ADMIN_PASSWORD = "YourStrongPass1"; py server.py --reset-demo --no-demo
```

> **Warning**
> `--reset-demo` permanently deletes the current database. Use it only before the system holds real data.

Server options

Command	What it does
`py server.py`	Normal start on port 8000.
`py server.py --port 8443`	Use a different port.
`py server.py --behind-proxy`	Use with Cloudflare Tunnel (internet access). Answers only on this PC; the tunnel serves HTTPS.
`py server.py --no-demo`	On an empty database, create only the admin account.
`py server.py --reset-demo`	Delete the database and recreate it.

5. Secure and configure the system

Sign in as `admin` and complete these steps in order. Each step uses items created in the previous one.

Change admin password → Settings → Departments → Locations → Shifts → Employees → Users

1. **Change the admin password**: My account → Change password. Passwords need at least 8 characters with a letter and a number. If you kept the demo data, also change or delete the demo accounts under Users.
2. **Settings**: enter the company name in Arabic and English, the time zone (minutes from UTC; `180` = Riyadh, UTC+3) and the attendance rules.
3. **Departments**: add your departments.
4. **Locations**: add each branch or site. Click the map to place the pin, or type the latitude and longitude, then set the Allowed radius (100–300 m suits most buildings).
5. **Shifts**: set start and end times, work days, grace periods, break and overtime threshold. An end time earlier than the start (e.g. 22:00 → 06:00) creates an overnight shift.
6. **Employees**: add each employee with a shift, a main Work location and, if they work at more than one branch, Additional work locations. Tick Must check in from inside the geofence to enforce GPS.
7. **Users**: create a login for each employee with the role *Employee* and link it to the employee record. Give HR staff and managers the matching roles.

> **Tip: employees at several branches**
> A punch is accepted when the employee is inside the radius of *any* of their sites (main + additional). The site used is saved with the punch.

6. Make it reachable from phones

Phones only allow GPS on a secure `https://` address. Choose one of three ways:

Method	Reach	Address	Best for
A. Local network certificate	Same Wi-Fi only	`https://192.168.x.x:8000` (certificate warning)	Testing in the office
B. Temporary tunnel	Internet, all branches	Random `https://….trycloudflare.com`, changes on every restart	Trials and demos
C. Permanent tunnel	Internet, all branches	Your own, e.g. `https://attendance.yourcompany.com`	**Daily use (recommended)**

A. Local network only

1. Double-click `gen_cert.bat`. It creates `certs\cert.pem` and `certs\key.pem`.
2. Restart the server. It now runs on `https://<PC-IP>:8000`; the address is shown in the console.
3. On each phone, open the address and accept the certificate warning.

> **Limitation**
> With this certificate, phones show a warning and offline mode may not work fully. Use method C for daily use.

B. Temporary internet address

1. Start the server in proxy mode: `py server.py --behind-proxy`
2. In a second PowerShell window, inside the folder, run:

```
powershell -ExecutionPolicy Bypass -File online.ps1
```

The script installs **cloudflared** if needed and prints an address like `https://xxxx-xxxx.trycloudflare.com`. Keep both windows open. The address changes every time the tunnel restarts.

C. Permanent internet address (recommended)

1. Create a free account at `cloudflare.com` and add your domain, following Cloudflare's steps to change the domain's name servers.
2. Open Zero Trust → Networks → Tunnels → Create a tunnel → Cloudflared. Name it, e.g. *attendance*.

3. Add a Public hostname: subdomain `attendance`, your domain, service type `HTTP`, URL `localhost:8000`.
4. Copy the tunnel **token** (the long text after `--token` in the install command Cloudflare shows).
5. Open PowerShell **as Administrator** in the system folder and run:

```
powershell -ExecutionPolicy Bypass -File online.ps1 -Token <your-token>
```

The tunnel is installed as a Windows service and starts with Windows. Then set the server to start automatically in proxy mode (chapter 7).

> **Before sharing an internet address**
> Make sure the demo passwords are changed (chapter 5). Anyone who has the address can reach the sign-in page.

7. Start automatically with Windows

`autostart.ps1` registers a hidden scheduled task that starts the server and restarts it within a minute if it stops.

Command (run in the system folder)	Effect
`powershell -ExecutionPolicy Bypass -File autostart.ps1 -BehindProxy`	Start at user sign-in, in proxy mode (for Cloudflare Tunnel).
`… autostart.ps1 -BehindProxy -AtBoot`	Start when Windows boots, before anyone signs in. Needs an Administrator PowerShell.
`… autostart.ps1`	Start at sign-in in normal mode (local network).
`… autostart.ps1 -Remove`	Remove the automatic start.

> **Power settings**
> In Windows Settings → Power, set the PC to **never sleep** while plugged in. A sleeping PC means nobody can check in.

8. Install the app on phones

Send employees the system address (e.g. `https://attendance.yourcompany.com`) with their username and first password.

Android (Chrome)

1. Open the address in **Chrome**.
2. Tap the menu ⋮ → Install app or Add to Home screen.

3. Open the app from the home screen, sign in and tap **Got it** on the location notice.
4. When asked, allow location access.

iPhone (Safari)

1. Open the address in **Safari**.
2. Tap **Share** ↑ → **Add to Home Screen**.
3. Open the app from the home screen and sign in.
4. Allow location when asked. If it was denied: **Settings** → **Privacy & Security** → **Location Services** → **Safari Websites** → *While Using*.

9. Backups, logs and restore

- A backup is created automatically once a day in `data\backups\`. The last 14 are kept; change the number in **Settings**.
- Admins can create and download a backup at any time from **Settings** → **Backup**.
- Server messages and errors are written to `data\server.log`.

Restore a backup

1. Stop the server: close its window, or stop the *Attendance Server* task in Task Scheduler.
2. Copy `data\attendance.db` somewhere safe, in case you need it again.
3. Delete `attendance.db-wal` and `attendance.db-shm` from `data\` if they exist.
4. Copy the chosen backup file to `data\attendance.db`.
5. Start the server again.

> **Off-site copy**
> Copy the `data\backups` folder to a USB drive or a cloud drive regularly. A backup on the same PC does not protect you if the PC fails.

10. Troubleshooting

Problem	Cause and solution
Phone says location requires a secure connection	The address is not `https://`. Use a tunnel (chapter 6, method B or C).
The temporary link stopped working	The tunnel or the PC was restarted, so the address changed. Run `online.ps1` again and share the new address, or set up the permanent tunnel.
The site does not open at all	Check that the PC is on and the server is running: open `http://localhost:8000` on the PC. Check the tunnel window or service.

Problem	Cause and solution
"Address already in use" on start	Another copy of the server is already running. Close it, or use `--port` with another number.
The new link does not open on the server PC only	Windows cached an old DNS answer. Run `ipconfig /flushdns`.
Employee rejected as "outside the allowed area"	Check the location pin and radius under **Locations**, and the employee's main and additional sites. GPS is less accurate inside large buildings, so a radius of 150–300 m may help.
Rejected for poor location accuracy	The phone's GPS fix is worse than **Settings** → maximum accuracy (150 m by default). Ask the employee to step outside or near a window.
Account locked	Five wrong passwords lock an account for 10 minutes. HR or the admin can reset the password under **Users**, which also unlocks it.
Employee sees "No shift is assigned to you"	Assign a shift on the employee record.
Employee sees "This account is not linked to an employee"	Edit the user under **Users** and choose the linked employee.

11. Security checklist

- ☐ Admin password changed; demo accounts removed or changed.
- ☐ Each person has their own account; no shared logins.
- ☐ HR and managers have only the roles they need.
- ☐ The system is reached only through HTTPS (tunnel).
- ☐ The server PC has Windows updates, antivirus and a password-protected Windows account.
- ☐ Backups are copied off the PC regularly.
- ☐ When an employee leaves: set their status to *Inactive* and disable their user account.
- ☐ Review the **Audit log** from time to time for unusual edits or failed logins.